3.1 Introduction

Once a general and targeted history, examination, and investigations have been carried out ('information gathering'), it is time for the dentist and their oral healthcare team to assimilate all of the relevant information in order to formulate a diagnosis, prognosis, and care plan for the individual patient ('information processing'). Although the detection and diagnostic phases are each discussed separately in this book, an experienced clinician will often accomplish both phases simultaneously. It is vital to remember that *diagnosis precedes treatment* in all cases.

3.1.1 Definitions

Diagnosis

Diagnosis is the art or act of inferring, from its signs and symptoms or manifestations, the nature or cause of an illness or condition. This stage is critical in order to allow the dental team and the patient to appreciate the nature, cause, and severity of the illness or condition.

Prognosis

The prognosis is the forecast of the course of a disease or the patient's response to treatment of the disease. This stage helps the dentist and the patient to understand how easy or difficult the treatment will be to carry out, and it allows assessment of the patient's motivation to cure the problem. In dentistry, the oral healthcare team can only start the patient off on the road to recovery by restoring form and function to their dentition as well as helping the patient to prevent or control the disease process, so preventing its return. It is then up to the patient whether they follow this advice and maintain their oral health in the future.

Patient-centred care plan

This plan is the formal itemized management strategy, developed by the dentist and their oral healthcare team, for the individual patient to treat the manifestations of a disease and to control it or prevent it from recurring. It can be divided into *phases* of therapy (e.g. prevention or control, stabilization or definitive treatment, and review, reassessment, or recall), and it should be adapted and modified during its execution for maximum benefit to the patient. It should take into account unforeseen developments in the course of the disease or the patient's response to care. It should be written down and made clear to all parties for discussion, so that informed consent can be gained prior to implementation.

3.2 Diagnosing dental pain, or 'toothache'

'Common diseases occur commonly.' A list of common diagnoses of dental pain follows, and the various features of each are summarized in Table 3.1.

- Acute pulpitis.
- Acute periapical periodontitis.
- Acute periapical abscess.
- Acute periodontal (lateral) abscess.
- Chronic pulpitis.
- Chronic periapical periodontitis (apical granuloma).
- Exposed sensitive dentine.
- Interproximal food packing.
- Cracked cusps/tooth syndrome.

Other conditions which are beyond the remit of this book and that may also manifest as dento-facial pain include maxillary sinusitis, trigeminal neuralgia, facial arthromyalgia, and pericoronitis. It must be understood that pain symptoms rarely fall neatly into the diagnoses listed previously (see also Table 3.1), and that a degree of clinical experience will be required to interpret the relevant signs and symptoms and thus permit a diagnosis to be made. It is important to recognize that the signs and symptoms from an inflamed pulp do not necessarily correlate fully with the histological changes that are occurring within that pulp. Therefore all interpretation of the signs and symptoms must be undertaken with caution.

3.2.1 Acute pulpitis

- Acute, severe, poorly localized pain.
- Does not cross the facial midline, and might be difficult for the patient to identify from which jaw it is originating. They may present holding their face on the relevant side, rather than identifying a particular tooth.
- There are two clinical presentations (not necessarily relating to pulp histology), *reversible* and *irreversible*. Table 3.2 highlights their similarities and differences.
- Clinically, offending teeth may present with carious lesions or large restorations which may be defective, but the dentist must always check the status of the dentition in both quadrants on the relevant side.

3.2.2 Acute periapical periodontitis

- Pain localizable to the offending tooth—the patient will be able to point to it and will complain of pain when biting on it (due to stimulation of pain- and pressure-sensitive fibres in the periodontal ligament).
- Tooth tender to palpation and percussion. This should only entail a gentle pushing action on suspect teeth into their sockets or laterally. It does not mean tapping the crown of the tooth with a mirror handle

Table 3.1 Classic signs and symptoms for differential diagnoses for dental pain (unfortunately, not all symptoms fall within these clear categories, and experience is required when interpreting them)

	Acute pulpitis	Acute apical periodontitis	Acute apical abscess	Acute periodontal abscess	Exposed sensitive dentine	Food packing	Cracked cusp	Chronic pulpitis	Chronic apical periodontitis (apical granuloma)
History	Recent pain with hot and cold stimuli. May be very severe. Poorly localized	Tender to bite. Well localized	Pain and swelling Very well localized	Localized swelling. Some pain	Generalized pain in response to hot, cold, and sweet stimuli	Pain after eating fibrous food, e.g. meat	Vague intermittent pain usually on biting. May be poorly localized	Vague, unprovoked intermittent but increasing pain. Poorly localized	May have had pain in the past. Now not sensitive to hot and cold
Clinical examination	Possibly caries or recent large restoration	Possibly caries	May be extra-oral or intra-oral swelling over apex of tooth	intra-oral swelling nearer to gingival margin. Tooth may be mobile	Gingival recession. Exposed dentine at the gingival margin. Sensitive to probe or cold air	Open contact points. Gingival inflammation. Food usually present	Often nothing, but crack may be seen. May be painful with occlusal contact only	May have large restoration or caries	May have large restoration or caries
Vitality test	Hypersensitive	May still be vital, but usually non-vital	Non-vital	Often vital	Vital	May be vital or non-vital	May be hypersensitive	Often normal but may be hypersensitive	Non-vital
Percussion	Not tender	Tender	Tender to touch. Too tender to percuss	Slight tenderness, more to lateral than axial pressure	Not tender	Not tender to percussion. May be sore with lateral percussion	Usually not tender but may be	Not tender	Slightly. May give dull sound on percussion
Other clinical tests			Raised temperature. Looks ill	Deep pockets. Pus may be released on probing pocket		Floss passes the contact easily	Sometimes tender to lateral pressure on an individual cusp		
Radiographic findings	Probably caries close to pulp. No periapical change	Usually no periapical change in early stage	Usually no periapical change except slight thickening of apical periodontal membrane	Alveolar bone loss. Usually no periapical change	May be some alveolar bone loss	None	None	None	Periapical radiolucency
Findings on further investigations	Carious exposure of pulp	Necrotic pulp	Pus may be drained via abscess cavity to root canal without local anaesthetic, giving immediate relief of pain and confirming the diagnosis				Crack sometimes visible at base of cavity when old restoration is removed. If left, cuspal fracture will eventually occur	Symptoms may settle if restoration is removed and tooth is dressed with calcium hydroxide, but pulp often dies eventually	Necrotic pulp

Table 3.2 Characteristics of acute pulpitis

Reversible pulpitis	Irreversible pulpitis
• Characteristic short, sharp pain • *Stimulated* by hot, cold, or sweet stimuli • *A few seconds*; *disappears* when stimulus is removed • Pulp sensibility tests may elicit an exaggerated response • Tooth is not tender to percussion (TTP)	• Characteristic dull, throbbing pain (although the patient may experience bouts of sharp pain) • Onset is usually unprovoked/*exacerbated* by hot, cold, or sweet stimuli • *Several minutes to several hours*; *persists* when stimulus is removed • Pulp sensibility tests may elicit an exaggerated or negative response • Tooth is not TTP (except in late-stage presentation)

(warn the patient that this examination may be uncomfortable, and obtain their permission before you undertake it).

- Tenderness felt over the root apex through mucosae in the buccal sulcus.
- Pulp may initially retain vitality, but will eventually become necrotic.
- Periapical radiograph *may* show loss of lamina dura around the root apex region during late-stage presentation.

3.2.3 Acute periapical abscess

- Patient may present with a large facial and/or intra-oral swelling associated with the affected relevant quadrant, or a more localized swelling in the alveolar mucosa over the affected root.
- May present before swelling occurs, or after it has burst and subsided, with evidence of a sinus tract.
- Patient may present with fever.
- Tooth will be painful to bite on; negative results with vitality tests (if permitted).
- Periapical radiograph will usually show loss of lamina dura and widening of the periodontal ligament space around the root apex.

3.2.4 Acute periodontal (lateral) abscess

- Forms at the base of a deep periodontal pocket.
- Well-localized pain associated with a vital pulp.
- If this is combined with pulpal necrosis (of differing aetiology), a 'perio-endo' lesion can be diagnosed, often with a poor prognosis, leading to extraction of the tooth.

3.2.5 Chronic pulpitis

- Mild, poorly localized, periodic, grumbling pain (on and off) over several weeks or months.
- Initially vital pulp and tooth not tender to percussion, but if condition progresses, symptoms of periapical periodontitis may eventually supervene.
- Initial treatment will involve monitoring to assess whether the pulp has recovered after treating the cause (e.g. caries, cracked tooth).
- More severe cases will require removal of any restoration, placement of a sedative dressing over the pulp, and then a provisional restoration.

3.2.6 Chronic periapical periodontitis

- Can be symptomless or may elicit mild pain on biting.
- Positive response to vigorous percussion, or even presentation with a sinus tract (see Figure 3.1).
- Periapical radiograph shows well-demarcated radiolucency around the root apex (see Figure 3.2).
- Often described as an apical granuloma.



Figure 3.1 **(a)** Periapical radiograph showing a well-demarcated radiolucency at the apex of LR5 (with accompanying loss of lamina dura and relative widening of the periodontal ligament space when compared with the adjacent healthy LR7). The large restoration has contributed to pulp necrosis in this tooth, and it is likely that a periapical granuloma has formed, but this should strictly be diagnosed from histological analysis of a biopsy sample. **(b)** Periapical radiograph of the same tooth after the gutta-percha root filling was placed. **(c)** Radiograph taken 3 years later, showing healing around the root apex of the LR5, bony infill, and re-formed lamina dura. (Courtesy of the late Professor Pitt Ford.)



Figure 3.2 Chronically infected apical granuloma **(a)** with pus pointing through a sinus on the mucosa adjacent to the non-vital LR6 **(b)**. **(c)** Periapical radiograph shows a large radiolucency associated with the apices and furcation of the LR6, as well as a separate periapical radiolucency on the heavily restored (but inadequately root-treated) LR5.

- The apical granuloma is a chronic inflammatory response (highly vascularized granulation tissue) to toxins leaching from the necrotic pulp and causing increased osteoclastic action in the subjacent bone, leading to bone resorption. As the toxins become diluted, the extent of the resorption is naturally limited. If the necrotic pulp, and therefore the source of toxins, is removed, bony repair can result (see Figure 3.2).
- If infection occurs, the chronic apical granuloma can flare up into an acute apical abscess with sinus tract (see Figure 3.1).
- Chronic apical granulomas may become cystic and require treatment by surgical intervention (see Figure 3.3).

3.2.7 Exposed sensitive dentine

- Poorly localized short, sharp sensitivity to hot, cold, or sweet stimuli, caused by exposed root dentine surfaces (gingival recession, root caries). The pain response can be elicited by air-drying the exposed surface using a dental 3-1 air/water syringe.
- Positive pulp test (perhaps slightly hyper-responsive if tested during an acute episode), and teeth not tender to percussion.
- To reduce the sensitivity, the exposed tubules may be blocked using a dentine bonding agent or a thin layer of resin composite. Use of potassium nitrate/chloride compounds may help to reduce tubule conduction and pain transmission.

3.2.8 Interproximal food packing

- Open contact point (drifted teeth, poor contouring of adjacent proximal restorations) enables food debris to get caught between teeth and traumatize the interdental gingival papillae, leading to localized tenderness, bleeding, and inflammation.

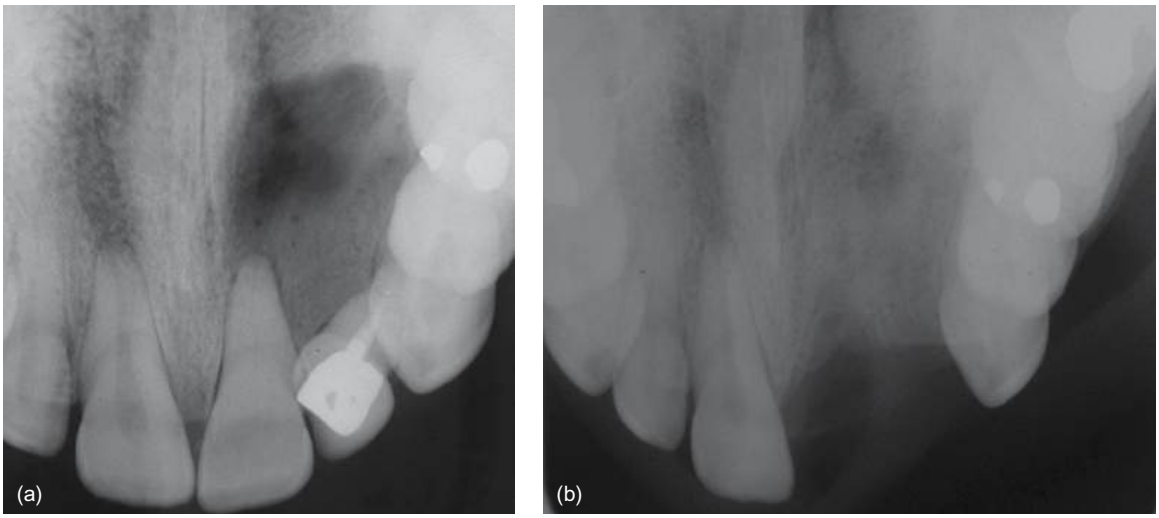


Figure 3.3 **(a)** Upper standard occlusal radiographs showing large palatal cystic change in an apical granuloma associated with UL12. **(b)** The same radiograph taken 3 years later after UL12 had been extracted and the cyst surgically managed. Note the near complete bony infill of the initially large defect. Implants were placed successfully into the new bone to restore UL12.

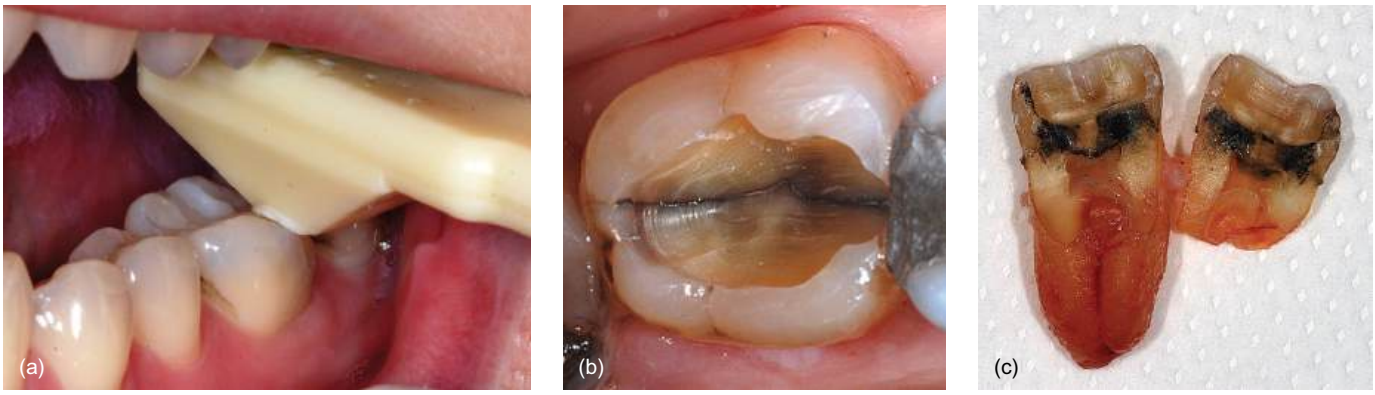


Figure 3.4 **(a)** A Tooth Slooth® (sleuth) detects whether there is a cracked cusp on the disto-buccal aspect of the left mandibular first molar. The patient bites on the plastic tip, which directs the occlusal force on to the disto-buccal cusp in this case. Pain may be elicited on pressure or on release as the cracked cusp flexes under occlusal strain. **(b)** A mandibular molar with symptoms of a cracked tooth. The large existing amalgam restoration was removed, revealing a clear fracture line mesio-distally (more easily detected using magnification; see Chapter 5, Section 5.7). As the portions of the tooth were mobile and the coronal extent of the fracture was considerable, the tooth was deemed unrestorable and it was extracted. **(c)** The subgingival extent of the oblique shear fracture, through the pulp chamber. (Courtesy of L. Mackenzie.)

3.2.9 Cracked cusp/tooth syndrome

- Cracks can involve enamel only (no pain, similar to superficial cracks in a porcelain teacup), or enamel and dentine (which may or may not include the pulp), which can cause poorly localized pain periodically.
- There is sometimes a sharp pain on biting/thermal stimulus. Pulp tests may be inconclusive, as may percussion and palpation tests.
- Transillumination or dyes can be used to detect cracks, but do not relate their presence to the cause of pain.
- Diagnostic testing may involve careful analysis of pain on biting. Using a cotton-wool roll, wood stick (wooden tongue depressor split lengthways), rubber suction tip, or proprietary kits (e.g. Tooth Slooth®), cracked tooth/cusp pain can be elicited on specific cusps, at specific angles of pressure, and assessed both on biting and release. This helps to ascertain where and how deep the crack might be, and thus what treatment is required (see Figure 3.4).

3.3 Caries risk/susceptibility assessment

The aetiology and histopathology of caries, as well as the methods for detecting it, have been discussed in Chapters 1 and 2. Once this information has been gathered through a targeted verbal history and examination, analysis ('information processing') is required to ascertain the risk and likelihood of the individual patient developing further disease and responding to treatment of current disease. This is achieved by relating the risk status of the patient on presentation to the activity state of any lesions present. This is known as a caries risk/susceptibility assessment or analysis. Without this knowledge, caries management at the individual patient level will never be successful in the long term. Risk assessments can also be carried out at a population level in order to help to promote public health prevention strategies for large numbers of people (e.g. water fluoridation programmes).

- It makes economic and practical sense to target preventive treatments in practice at the appropriate risk group.
- Dental care neither begins nor ends with a single course of treatment, but is ongoing. When a course of dental treatment is complete, the dentist, team, and patient decide when it would be wise to check that all is well. This recall interval is based partly on an assessment of the likelihood and risk of disease progression, and should not be standardized at 6 months or any other period.
- The patient must be made aware of their risk status. This knowledge encourages them to attend appropriately timed recall appointments, to become motivated and involved, taking responsibility for their own preventive care, and, if they pay for their treatment, it may help them to budget for dental bills.

The various patient factors that affect the risk assessment have been highlighted in Table 3.3. The relevance of these factors in a targeted verbal history of risk assessment has been indicated in Table 2.1 (in Chapter 2), and a chart similar to Table 3.4 might be used in the patient notes to document such findings. Formal, 'standardized' risk assessment systems exist which help the oral healthcare team to calculate their patients' risk status (e.g. Cariogram, Caries Management by Risk Assessment (CAMBRA), PreViser™; QR codes images 3.1 and 3.2 will take you to further information on these systems). In some cases, algorithms are used to process the findings from the patient assessment, giving extra weight and relevance to certain factors that are deemed more important to the risk outcome. Although these can be convenient and helpful



QR Code image 3.1 Scan this code with your mobile device to view the Cariogram risk assessment system.

www.mah.se/fakulteter-och-omraden/Odontologiska-fakulteten/Avdelning-och-kansli/Cariologi/Cariogram



QR Code image 3.2 Scan this code with your mobile device to view the PreViser™ risk assessment system.

www.previser.co.uk

for teaching and research purposes, there is a risk that the analysis will become too standardized and the nuances and variations for each patient will not be fully appreciated or understood, especially at general practice level. It is more important for the dentist and the oral health-care team to tailor the risk assessment for each patient, with the team and the patient appreciating and understanding fully the relevance and significance of all the information gathered for that specific individual. With the increase in litigation against dentists, risk assessments that are interpreted and recorded formally in the clinical notes are becoming essential dento-legal requirements, not forgetting the obvious ethical

Table 3.3 Factors involved in assessing caries risk and categorizing the patient's susceptibility to caries as high or low risk*

Caries risk assessment		
High risk		**Low risk**
Social history	Socially deprived High caries in siblings Low knowledge of dental disease Irregular attender Ready availability of snacks Low dental aspirations	Middle class Low caries in siblings Dentally aware Regular attender Work does not allow regular snacks High dental aspirations
Medical history	Medically compromised Disabled Xerostomia Long-term cariogenic medicine	No medical problem No physical problem Normal salivary flow No long-term medication
Dietary habits	Frequent sugar intake	Infrequent sugar intake
Fluoride use	Non-fluoride area No fluoride supplements No fluoride toothpaste	Fluoridation area Fluoride supplements used Fluoride toothpaste used
Plaque control	Infrequent, ineffective cleaning Poor manual control	Frequent/eff ective cleaning Good manual control
Saliva	Low flow rate Low buff ering capacity High *Streptococcus mutans* and Lactobacillus counts	Normal flow rate High buffering capacity Low *S. mutans* and lactobacillus counts
Clinical evidence	New lesions Premature extractions Anterior caries or restorations Multiple restorations History of repeated restorations No fissure sealants Multiband orthodontics Partial dentures	No new lesions Nil extractions for caries Sound anterior teeth No or few restorations Restorations inserted years ago Fissure sealed No appliances

*This information can be gathered from a targeted verbal history and examination (see Chapter 2).

Table 3.4 An example of a chart for noting down key findings from the patient's history and examination in order to help to evaluate their caries risk

Status	'Yes' answer: high risk	'No' answer: low risk
Lesions: Two or more new, progressing, and/or restored lesions in the last 2 years? Activity state?		
General factors		
Diet: Frequent snacks between meals? Anorexia or bulimia? Prolonged nursing/ bottle feeding?		
Fluoride: Deficient fluoride exposure (toothpaste/rinse daily, community water fluoridation)?		
Health: Sjögren's syndrome, chemotherapy, radiation to head and neck?		
Medications: Hypo-salivatory medication, polypharmacy?		
Social: Socio-economic status? Parents/siblings with active caries? Symptom-driven attendance?		
Age: Adolescent? Elderly?		
Oral factors		
Oral hygiene: Quality? What aids are used?		
Saliva: Stimulated saliva flow < 0.7 mL/min? pH/buffering capacity?		
Plaque: Readily visible heavy plaque—plaque scores?		
Bacterial balance: Levels of *Streptococcus mutans*		

First published in the *Journal of Minimum Intervention Dentistry* 2009; 2: 103–24. Reproduced here with permission.

dimension to this best practice. The caries risk assessment will change over time, and should therefore be undertaken when the pattern of disease presentation changes in the same patient—the overall patient management of caries is as dynamic as the disease process itself.

With regard to the number of lesions detected during the clinical oral examination, a patient at high risk for caries is one who presents with or has a history of two or more new, progressing, and/or restored carious lesions in the previous 2 years. The patient's caries risk is classified generally into three categories—high, medium, or low (or red, amber, or green (RAG) traffic lights). Although this is useful for epidemiological and research purposes, it can add an unnecessary layer of complexity, or even ambiguity, for the oral healthcare team when assessing patients in general dental practice. A simpler division into low or high risk/susceptibility is usually more than sufficient to paint the individual risk picture for the patient, and aids the formulation of their individualized care plan accordingly. It is of course recognized that there are patient subgroups within the general population who are susceptible to certain factors (e.g. radiotherapy, autoimmune conditions) that elevate their risk to an extreme level (also classified as high risk with unmodifiable factors), and these individuals may require periodic referral for advice about care planning, and indeed may sometimes need care in a specialist environment (see Chapter 4, Section 4.2.1).

3.4 Diagnosing tooth wear

The detection and diagnosis of tooth wear from its causative factors have been discussed in detail in Chapter 2. A targeted verbal history coupled with rigorous examination of the teeth is essential for reaching a diagnosis of this common multifactorial problem. Differentiating between acceptable age-related and pathological levels of wear can be difficult, as this depends on the patient's age, dental history, and

the rate of progress of the tooth wear, which can be difficult to assess clinically unless the patient is reviewed annually in the first instance. However, wear facets with staining or dental plaque present may be indicative of a very slow or arrested rate of progress, whereas significant dental sensitivity from such lesions may indicate a more rapid rate of wear, as there would not have been enough time for the up-regulated odontoblasts to lay down translucent or tertiary dentine (see Chapter 1, Section 1.2.8).

3.5 Diagnosing dental trauma and developmental defects

Again, as with tooth wear, the degree of separation between detection and diagnosis of trauma and developmental defects is negligible, as the two processes are linked intimately with the verbal history and the oral/dental examination. These have been discussed in detail in Chapter 2.

3.6 Prognostic indicators

The dentist should give the patient a verbal and written prognosis of their responsibilities and the potential outcomes of preventive care, the individual tooth or teeth requiring interventional treatment, and/or a prognosis for the management of the overall condition. At the tooth level, factors that affect the prognosis include:

- *Restorability*—that is, the severity of damage incurred to the coronal hard tissues (extent of the carious lesion, and amount of tooth structure lost due to trauma, tooth wear, or developmental defect). Can operative treatment be used to repair and reinstate the tooth structure successfully in the mid to long term?
- The status of the pulp.
- The status of the periodontium (alveolar bone levels, periodontal ligament).

With regard to the underlying condition, prognostic factors include:

- The ease with which the aetiological factors can be removed or modified to prevent or slow the progress of the disease and its recurrence. For example, the aetiology of xerostomia caused by radiation therapy cannot be modified, so dental conditions (e.g. caries) will generally have a poor prognosis. However, with the judicious use of oral hygiene procedures to a high standard, complemented with high concentrations of fluoride, the prognosis may be improved.
- The overall level of patient motivation to manage the condition. If the patient is not particularly concerned or does not wish to take responsibility for maintaining their oral health, the long-term outlook for treatment administered by the oral healthcare team will be bleak at best, and only repairs can be considered. This, together with the long-term consequences of further tooth destruction, must be made clear to the patient and documented fully in the clinical notes.

3.7 Formulating an individualized care plan

3.7.1 Why is a care plan necessary?

The need for and importance of making individualized, adaptable holistic plans (both simple and complex), and of recording the decisions in the patient's clinical record, can be summarized as follows:

- It ensures that the lead clinician reviews the care in the light of all available evidence at the start of treatment and at intervals throughout treatment.
- It is a record for later reference, particularly in complicated cases and after a lapse of time during the treatment period. This is available primarily for the benefit of the patient, but on occasion, when a patient makes a complaint, it will have dento-legal importance and may protect the dentist and their team against unjustified complaints. In this context, when the care plan is complex and expensive, it is essential to put the advice in the form of a letter to the patient so that he or she has time to consider it, is able to digest it and its implications, and has an opportunity to question it before accepting or rejecting it. This process of two-way communication between the dentist and their team and the patient is of critical importance. The vast majority (more than 90%) of all dento-legal litigation pursued in the UK in 2013 was blamed on poor communication or understanding between the two parties, lack of management of patients' expectations, or inadequate record-keeping. In the UK the General Dental Council now suggests that this formalized two-way approach to planning is recommended practice in all cases.
- It avoids the risk of disorderly and ill-advised treatment, which may occur if treatment is administered piecemeal or by different members of the oral healthcare team or different specialists.

Written care plans, however simple, should be drawn up for every patient. They should be tailored specifically to the individual patient's holistic needs, must take all factors into consideration from the detection and diagnostic phases, and must have realistic goals and be achievable for the dentist, the oral healthcare team, and the patient. They must also be planned by the dentist and their team in such a way as to manage patient expectations with regard to the quality and success of any potential outcome of care. They must include all aspects of care, including non-operative preventive care and recall strategy, rather than just being a list of specific surgical procedures performed by the dentist to treat symptoms

of the disease. This is why the traditional 'treatment-plan' terminology is dated and inappropriate clinically. All too often such treatment plans became a shopping list of procedures to be carried out, rather than outlining—with reasons—the purpose of the different phases of care and how they interlink with the responsibilities of the oral healthcare team, the dentist, and the patient to secure a successful outcome.

Patients must understand the implications of the treatment and appreciate that the care plan may deviate from the initial path as the care progresses. They must also acknowledge their own pivotal role in its ultimate success, the time frame for its delivery, and its cost (if applicable).

3.7.2 Structure of the care plan

However simple a care plan may be for a particular management episode, it is wise to have a structure to follow. One such scheme is outlined in Table 3.5, and consists of three phases:

- an initial *stabilization phase* (which may take several months to complete, depending on the amount and type of work required)
- a *reassessment phase*—to re-evaluate the response to the initial treatment, and the future needs of the patient (which may have changed since the time of presentation)
- a *rehabilitation phase*—in which definitive restorations can be designed and placed.

Other considerations include the type of restorative material that is to be used (see Chapter 7), and whether operative intervention will be minimally invasive or more extensive (i.e. whether to restore or attempt to arrest a carious lesion in dentine, restore or monitor an erosive tooth-wear lesion, or extract or endodontically treat an infected tooth). These decisions must be made with input from the patient (including their own perceived need, aesthetic considerations, the number of appointments necessary, costs, and the level of dental anxiety, i.e. appreciating and managing the patient's expectations) as this will confer ownership and therefore responsibility to the patient in terms of the success of the final outcome. Second opinions from more experienced specialists (either practice or hospital based) can be very helpful, especially for the more complex cases, as there is rarely a single right or wrong decision to be made in care planning. Choosing the best of several options for the patient is a skill that develops with further clinical experience, and unfortunately this will involve making some errors of judgement along the way! It is interesting to note that these 'errors of judgement' rarely give rise to formal dento-legal action by patients, so long as the patient has been involved in the journey and the decision making. Good communication and documentation is the key. Always be prepared to *justify* the successful care plan to three parties—the patient, yourself, and a lawyer! It is vital to learn from these situations and adapt your own evidence base accordingly in the future.

Table 3.5 A care plan flowchart that structures patient care into three phases, which can blend into one another depending on the complexity of the plan.

Care plan phase	Comments
Stabilization	Pain relief ↓ Control active disease Prevention Extraction of teeth with hopeless prognosis Temporary/transitional restorations ↓
Reassessment	Reassess the patient's response to the stabilization phase, their subsequent needs, aesthetic expectations, motivation, dental examination, and costs ↓
Rehabilitation	Reorganizing or conforming to existing occlusal scheme (see later); definitive restorations/prostheses ↓ Review, reassess, and recall (maintenance)

Table 4.1 (in Chapter 4) gives an example of a care plan flowchart for patients with dental caries. It is based on five possible management options, dependent on the presence or absence of active lesions coupled with the individual's caries risk assessment (the overall caries matrix). Of course, real life is not always that simple, and patients' conditions do not always fit within such clear-cut categories, but this concept mapping exercise helps the dentist and the patient to understand the options and where the chosen management strategy fits in relation to the alternatives.

3.8 PubMed keywords



<www.ncbi.nlm.nih.gov/pubmed/advanced>

QR code image 3.3 Try searching the following keywords on PubMed for relevant further reading. You can access PubMed by scanning the QR code image or at the address

Keywords

Basic Erosive Wear Examination (BEWE); Smith and Knight Tooth Wear Index.